In [2]:
```python
# Q#1 - Student Result Management System
def calculate_result(student):
    total = (student["python"]+ student["ai"]+ student["math"])
    percentage = (total / 300) * 100
    if percentage >= 80:
        grade = "A"
    elif percentage >= 70:
        grade = "B"
    elif percentage >= 60:
        grade = "C"
    elif percentage >= 50:
        grade = "D"
    else:
        grade = "F"
    return total, percentage, grade
students = []

for i in range(5):
    print("\nEnter details for Student", i + 1)
    name = input("Enter student name: ")
    roll_no = input("Enter roll number: ")
    python_marks = float(input("Enter Python marks: "))
    ai_marks = float(input("Enter Artificial Intelligence marks: "))
    math_marks = float(input("Enter Mathematics marks: "))
    student = {
    "name": name,
    "roll_no": roll_no,
    "python": python_marks,
    "ai": ai_marks,
    "math": math_marks
 }
    students.append(student)

print("\n--- Student Results ---")
highest_student = None
highest_percentage = -1
for student in students:
    total, percentage, grade = calculate_result(student)
    print("\nName:", student["name"])
    print("Roll No:", student["roll_no"])
    print("Total Marks:", total)
    print("Percentage:", percentage)
    print("Grade:", grade)
    if percentage > highest_percentage:
        highest_percentage = percentage
        highest_student = student

highest_total, highest_percentage, highest_grade = calculate_result(highest_student
print("\n--- Highest Percentage ---")
print("Name:", highest_student["name"])
print("Roll No:", highest_student["roll_no"])
print("Percentage:", highest_percentage)
print("Grade:", highest_grade)
```

```
with open("results.txt", "w") as file:
    file.write("--- Student Results ---\n")
    for student in students:
        total, percentage, grade = calculate_result(student)
        file.write("\nName: " + student["name"] + "\n")
        file.write("Roll No: " + student["roll_no"] + "\n")
        file.write("Total Marks: " + str(total) + "\n")
        file.write("Percentage: " + str(percentage) + "\n")
        file.write("Grade: " + grade + "\n")
        file.write("\n--- Highest Percentage ---\n")
        file.write("Name: " + highest_student["name"] + "\n")
        file.write("Roll No: " + highest_student["roll_no"] + "\n")
        file.write("Percentage: " + str(highest_percentage) + "\n")
        file.write("Grade: " + highest_grade + "\n")
print("\nResults saved successfully in results.txt")
```

```
Enter details for Student 1
Enter details for Student 2
Enter details for Student 3
Enter details for Student 4
Enter details for Student 5
```

```
--- Student Results ---

Name: Amna
Roll No: 23
Total Marks: 240.0
Percentage: 80.0
Grade: A

Name: Eman
Roll No: 30
Total Marks: 270.0
Percentage: 90.0
Grade: A

Name: Zara
Roll No: 99
Total Marks: 254.0
Percentage: 84.66666666666667
Grade: A

Name: Fatima
Roll No: 45
Total Marks: 150.0
Percentage: 50.0
Grade: D

Name: Farah
Roll No: 24
Total Marks: 90.0
Percentage: 30.0
Grade: F

--- Highest Percentage ---
Name: Eman
Roll No: 30
Percentage: 90.0
Grade: A

Results saved successfully in results.txt
```

In [3]:

```python
# Q#2 - Sentence Analysis Program
def count_vowels(sentence):
    vowels = "aeiouAEIOU"
    count = 0
    for character in sentence:
        if character in vowels:
            count = count + 1
    return count
def analyze_sentence(sentence):
    words = sentence.split()
    total_words = len(words)
    total_characters = len(sentence)
    longest_word = max(words, key=len)
    shortest_word = min(words, key=len)
    vowels = count_vowels(sentence)
    unique_words = set(words)
```

```
    alphabetical_words = sorted(unique_words)
    return (total_words,total_characters,longest_word,shortest_word,vowels,unique_w
sentence = input("Enter a sentence: ")
(total_words,total_characters,longest_word,shortest_word,vowels,unique_words,alphab
print("\n--- Sentence Analysis ---")
print("Total words:", total_words)
print("Total characters:", total_characters)
print("Longest word:", longest_word)
print("Shortest word:", shortest_word)
print("Number of vowels:", vowels)
print("Unique words:", unique_words)
print("Words in alphabetical order:")
for word in alphabetical_words:
    print(word)
with open("word_analysis.txt", "w") as file:
    file.write("--- Sentence Analysis ---\n")
    file.write("Total words: " + str(total_words) + "\n")
    file.write("Total characters: " + str(total_characters) + "\n")
    file.write("Longest word: " + longest_word + "\n")
    file.write("Shortest word: " + shortest_word + "\n")
    file.write("Number of vowels: " + str(vowels) + "\n")
    file.write("Unique words: " + str(unique_words) + "\n")
    file.write("Words in alphabetical order:\n")
    for word in alphabetical_words:
        file.write(word + "\n")
print("\nAnalysis saved successfully in word_analysis.txt")
```

```
--- Sentence Analysis ---
Total words: 3
Total characters: 13
Longest word: Python
Shortest word: I
Number of vowels: 4
Unique words: {'I', 'Python', 'Love'}
Words in alphabetical order:
I
Love
Python

Analysis saved successfully in word_analysis.txt
```

In [5]:

```
# Q#3 - Simple Shopping Program

def calculate_bill(products):
    bill = 0
    for product in products:
        bill = bill + product["total"]
    if bill > 10000:
        discount = bill * 0.10
        final_bill = bill - discount
    else:
        discount = 0
        final_bill = bill
    return bill, discount, final_bill

products = []
```

```
for i in range(5):
    print("\nEnter details for Product", i + 1)
    name = input("Enter product name: ")
    price = float(input("Enter price: "))
    quantity = int(input("Enter quantity: "))

    total = price * quantity

    product = {
    "name": name,
    "price": price,
    "quantity": quantity,
    "total": total
 }

    products.append(product)

bill, discount, final_bill = calculate_bill(products)

most_expensive = products[0]
for product in products:
    if product["price"] > most_expensive["price"]:
        most_expensive= product

print("\n--- Shopping Bill ---")
for product in products:
        print(product["name"],"x",product["quantity"],"=",product["total"])
print("\nTotal Bill:", bill)
print("Discount:", discount)
print("Final Bill:", final_bill)
print("\nMost Expensive Product:", most_expensive["name"])
print("Price:", most_expensive["price"])

with open("bill.txt", "w") as file:
    file.write("--- Shopping Bill ---\n")
    for product in products:
        file.write(product["name"] + " x " +str(product["quantity"]) + " = " +str(p
    file.write("\nTotal Bill: " + str(bill) + "\n")
    file.write("Discount: " + str(discount) + "\n")
    file.write("Final Bill: " + str(final_bill) + "\n")
    file.write("\nMost Expensive Product: " +most_expensive["name"] + "\n")
    file.write("Price: " +str(most_expensive["price"]) + "\n")
print("\nBill saved successfully in bill.txt")
```

```
Enter details for Product 1
Enter details for Product 2
Enter details for Product 3
Enter details for Product 4
Enter details for Product 5
```

```
--- Shopping Bill ---
milk bottle x 2 = 200.0
butter box x 1 = 300.0
juice can x 1 = 300.0
cold drink can x 2 = 800.0
biscuit packets x 50 = 200.0

Total Bill: 1800.0
Discount: 0
Final Bill: 1800.0

Most Expensive Product: cold drink can
Price: 400.0

Bill saved successfully in bill.txt
```

In [8]:
```python
# Q#4 - Simple Calculator

def add(a, b):
    return a + b
def subtract(a, b):
    return a - b
def multiply(a, b):
    return a * b
def divide(a, b):
    return a / b
def modulus(a, b):
    return a % b

while True:
    num1 = float(input("Enter first number: "))
    num2 = float(input("Enter second number: "))
    operation = input("Enter operation (+, -, *, /, %): ")
    if operation == "+":
        result = add(num1, num2)
    elif operation == "-":
        result = subtract(num1, num2)
    elif operation == "*":
        result = multiply(num1, num2)
    elif operation == "/":
        result = divide(num1, num2)
    elif operation == "%":
        result = modulus(num1, num2)
    else:
        print("Invalid operation")
        continue
    print("Result:",result)
    again = input("Do you want another calculation? (yes/no): ")
    if again.lower() == "no":
        break
print("Calculator closed.")
```

```
Result: 47.0
Result: -33.0
Result: 2070.0
Result: 45.0
```

```
Calculator closed.
```

In [10]:
```python
# Q#5 - Simple Restaurant Program
menu = {
"Burger": 500,
"Pizza": 800,
"Fries": 250,
"Drink": 150
}
def calculate_total(price, quantity):
    return price * quantity
total_bill = 0
orders = []
while True:
    print("\n--- Restaurant Menu ---")
    print("1. Burger - Rs. 500")
    print("2. Pizza - Rs. 800")
    print("3. Fries - Rs. 250")
    print("4. Drink - Rs. 150")
    item = input("Enter item name: ")
    if item in menu:
        quantity = int(input("Enter quantity: "))
        price = menu[item]
        total = calculate_total(price, quantity)
        total_bill = total_bill + total
        order = {
        "item": item,
        "price": price,
        "quantity": quantity,
        "total": total
       }
        orders.append(order)
        print("Item total:", total)
        again = input("Do you want to order another item? (yes/no): ")
        if again.lower() == "no":
            break
    else:
        print("Item not available.")
print("\n--- Final Bill ---")
for order in orders:
    print(order["item"],"x",order["quantity"],"=",order["total"])
    print("Final Bill:",total_bill)
with open("orders.txt", "w") as file:
    file.write("--- Restaurant Bill ---\n")
    for order in orders:
        file.write(order["item"] + " x " +str(order["quantity"]) + " = " +str(order
    file.write("\nFinal Bill: " + str(total_bill))
print("Bill saved successfully in orders.txt")
```

```
--- Restaurant Menu ---
1. Burger - Rs. 500
2. Pizza - Rs. 800
3. Fries - Rs. 250
4. Drink - Rs. 150
Item total: 1000
```

```
--- Restaurant Menu ---
1. Burger - Rs. 500
2. Pizza - Rs. 800
3. Fries - Rs. 250
4. Drink - Rs. 150
Item not available.

--- Restaurant Menu ---
1. Burger - Rs. 500
2. Pizza - Rs. 800
3. Fries - Rs. 250
4. Drink - Rs. 150
Item not available.

--- Restaurant Menu ---
1. Burger - Rs. 500
2. Pizza - Rs. 800
3. Fries - Rs. 250
4. Drink - Rs. 150
Item not available.

--- Restaurant Menu ---
1. Burger - Rs. 500
2. Pizza - Rs. 800
3. Fries - Rs. 250
4. Drink - Rs. 150
Item total: 1600
--- Final Bill ---
Burger x 2 = 1000
Final Bill: 2600
Pizza x 2 = 1600
Final Bill: 2600
Bill saved successfully in orders.txt
```

In [ ]: